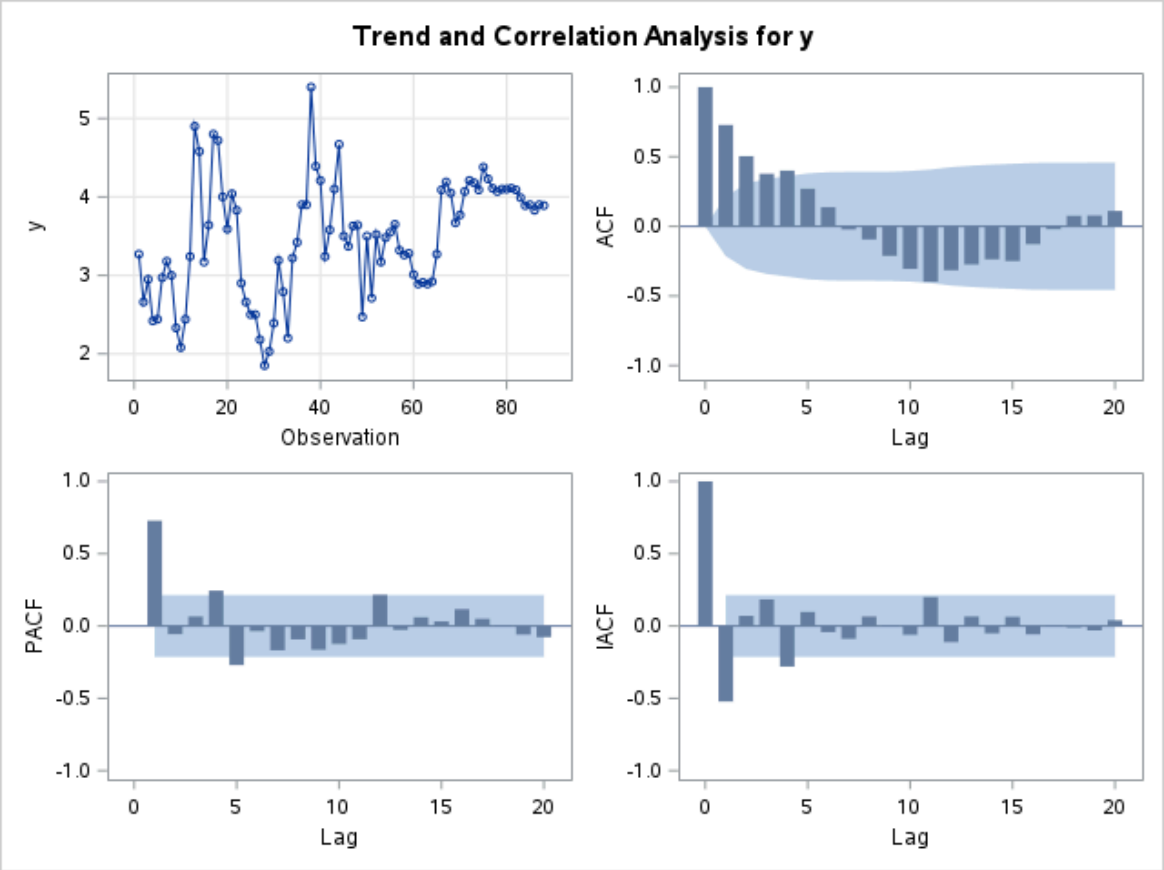
The ARIMA Procedure

Name of Variable = y	
Mean of Working Series	3.480909
Standard Deviation	0.73056
Number of Observations	88

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	108.85	6	<.0001	0.728	0.504	0.378	0.400	0.270	0.137
12	150.42	12	<.0001	-0.022	-0.094	-0.212	-0.305	-0.398	-0.315
18	173.59	18	<.0001	-0.274	-0.237	-0.249	-0.128	-0.020	0.076



Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	3.47082	0.18703	18.56	<.0001	0
AR1,1	0.73101	0.07388	9.89	<.0001	1

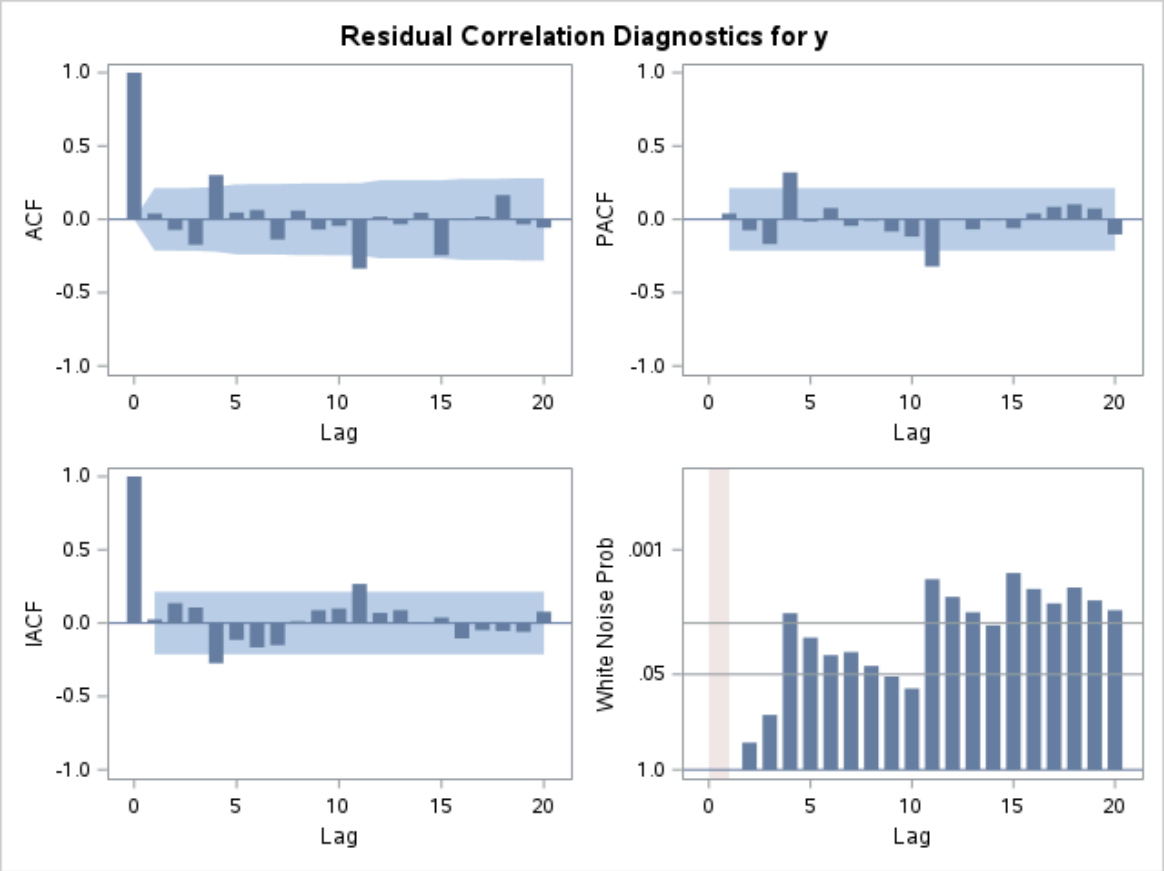
Constant Estimate	0.933633
Variance Estimate	0.255431
Std Error Estimate	0.505402
AIC	131.6075
SBC	136.5621
Number of Residuals	88

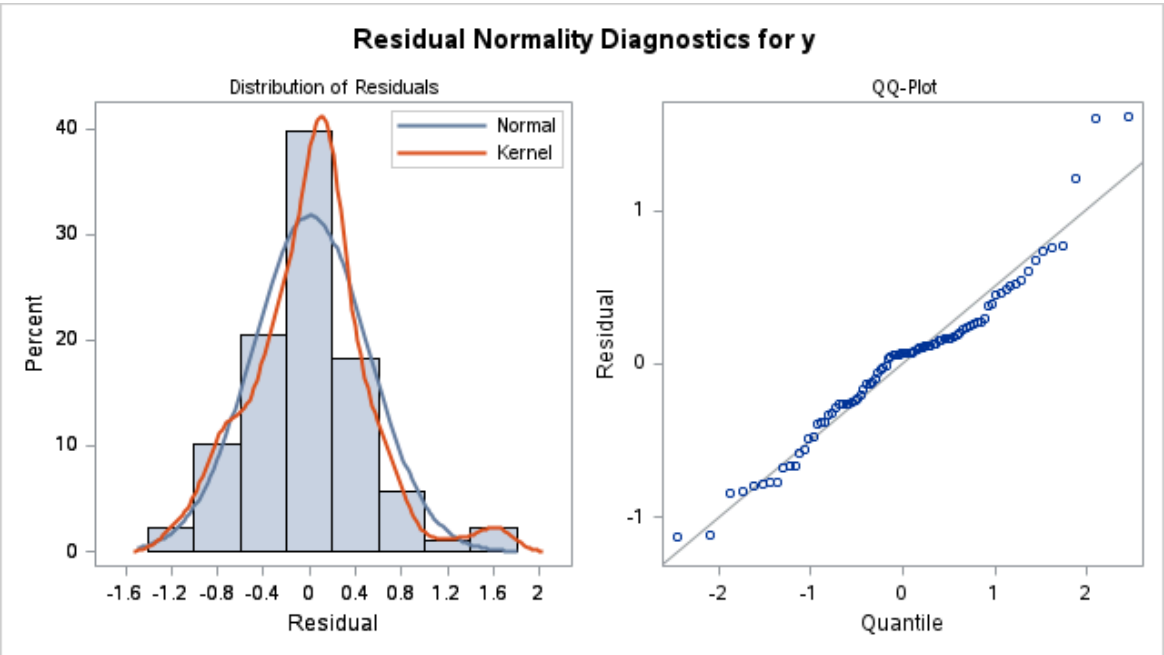
*** AIC and SBC do not include log determinant.**

Correlations of Parameter Estimates		
Parameter	MU	AR1,1
MU	1.000	-0.006

Correlations of Parameter Estimates		
Parameter	MU	AR1,1
AR1,1	-0.006	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	12.62	5	0.0272	0.040	-0.073	-0.173	0.302	0.047	0.063
12	27.08	11	0.0045	-0.138	0.060	-0.069	-0.045	-0.335	0.018
18	37.00	17	0.0034	-0.034	0.045	-0.245	0.004	0.019	0.164
24	40.35	23	0.0140	-0.035	-0.057	0.129	0.076	0.029	0.022

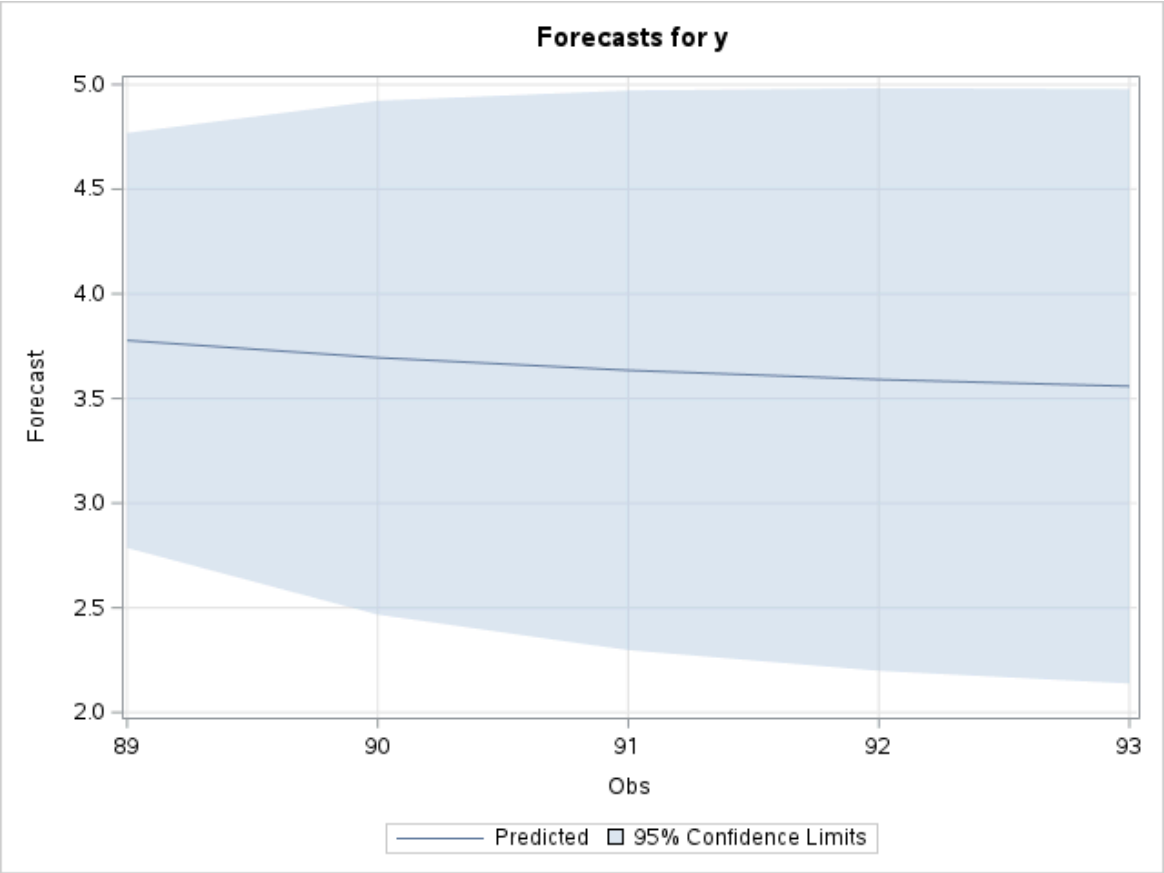




Model for variable y	
Estimated Mean	3.470818

Autoregressive Factors	
Factor 1:	1 - 0.73101 B**(1)

Forecasts for variable y				
Obs	Forecast	Std Error	95% Confidence Limits	
89	3.7772	0.5054	2.7867	4.7678
90	3.6948	0.6260	2.4678	4.9218
91	3.6346	0.6818	2.2982	4.9709
92	3.5905	0.7098	2.1993	4.9817
93	3.5583	0.7243	2.1386	4.9780



Obs	y	ylag1	ylag2	t
1	3.27	.	.	1
2	2.66	3.27	.	2
3	2.95	2.66	3.27	3
4	2.42	2.95	2.66	4
5	2.44	2.42	2.95	5
6	2.97	2.44	2.42	6
7	3.18	2.97	2.44	7
8	3.00	3.18	2.97	8
9	2.33	3.00	3.18	9
10	2.08	2.33	3.00	10
11	2.44	2.08	2.33	11
12	3.24	2.44	2.08	12

Obs	y	ylag1	ylag2	t
13	4.90	3.24	2.44	13
14	4.58	4.90	3.24	14
15	3.17	4.58	4.90	15
16	3.64	3.17	4.58	16
17	4.80	3.64	3.17	17
18	4.72	4.80	3.64	18
19	4.00	4.72	4.80	19
20	3.59	4.00	4.72	20
21	4.04	3.59	4.00	21
22	3.83	4.04	3.59	22
23	2.90	3.83	4.04	23
24	2.66	2.90	3.83	24
25	2.50	2.66	2.90	25
26	2.50	2.50	2.66	26
27	2.18	2.50	2.50	27
28	1.85	2.18	2.50	28
29	2.03	1.85	2.18	29
30	2.39	2.03	1.85	30
31	3.19	2.39	2.03	31
32	2.79	3.19	2.39	32
33	2.20	2.79	3.19	33
34	3.22	2.20	2.79	34
35	3.42	3.22	2.20	35
36	3.90	3.42	3.22	36
37	3.90	3.90	3.42	37
38	5.40	3.90	3.90	38
39	4.39	5.40	3.90	39
40	4.21	4.39	5.40	40
41	3.24	4.21	4.39	41
42	3.58	3.24	4.21	42
43	4.10	3.58	3.24	43
44	4.67	4.10	3.58	44
45	3.50	4.67	4.10	45
46	3.37	3.50	4.67	46
47	3.63	3.37	3.50	47
48	3.64	3.63	3.37	48
49	2.47	3.64	3.63	49
50	3.50	2.47	3.64	50
51	2.71	3.50	2.47	51
52	3.52	2.71	3.50	52
53	3.17	3.52	2.71	53
54	3.48	3.17	3.52	54
55	3.55	3.48	3.17	55
56	3.65	3.55	3.48	56
57	3.32	3.65	3.55	57
58	3.26	3.32	3.65	58
59	3.28	3.26	3.32	59
60	3.01	3.28	3.26	60
61	2.89	3.01	3.28	61
62	2.91	2.89	3.01	62
63	2.89	2.91	2.89	63
64	2.92	2.89	2.91	64
65	3.27	2.92	2.89	65
66	4.09	3.27	2.92	66

Obs	y	ylag1	ylag2	t
67	4.19	4.09	3.27	67
68	4.05	4.19	4.09	68
69	3.67	4.05	4.19	69
70	3.77	3.67	4.05	70
71	4.07	3.77	3.67	71
72	4.21	4.07	3.77	72
73	4.18	4.21	4.07	73
74	4.09	4.18	4.21	74
75	4.38	4.09	4.18	75
76	4.23	4.38	4.09	76
77	4.11	4.23	4.38	77
78	4.07	4.11	4.23	78
79	4.10	4.07	4.11	79
80	4.10	4.10	4.07	80
81	4.11	4.10	4.10	81
82	4.09	4.11	4.10	82
83	3.99	4.09	4.11	83
84	3.89	3.99	4.09	84
85	3.90	3.89	3.99	85
86	3.83	3.90	3.89	86
87	3.90	3.83	3.90	87
88	3.89	3.90	3.83	88

data from oil.dat

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	88
Number of Observations Used	87
Number of Observations with Missing Values	1

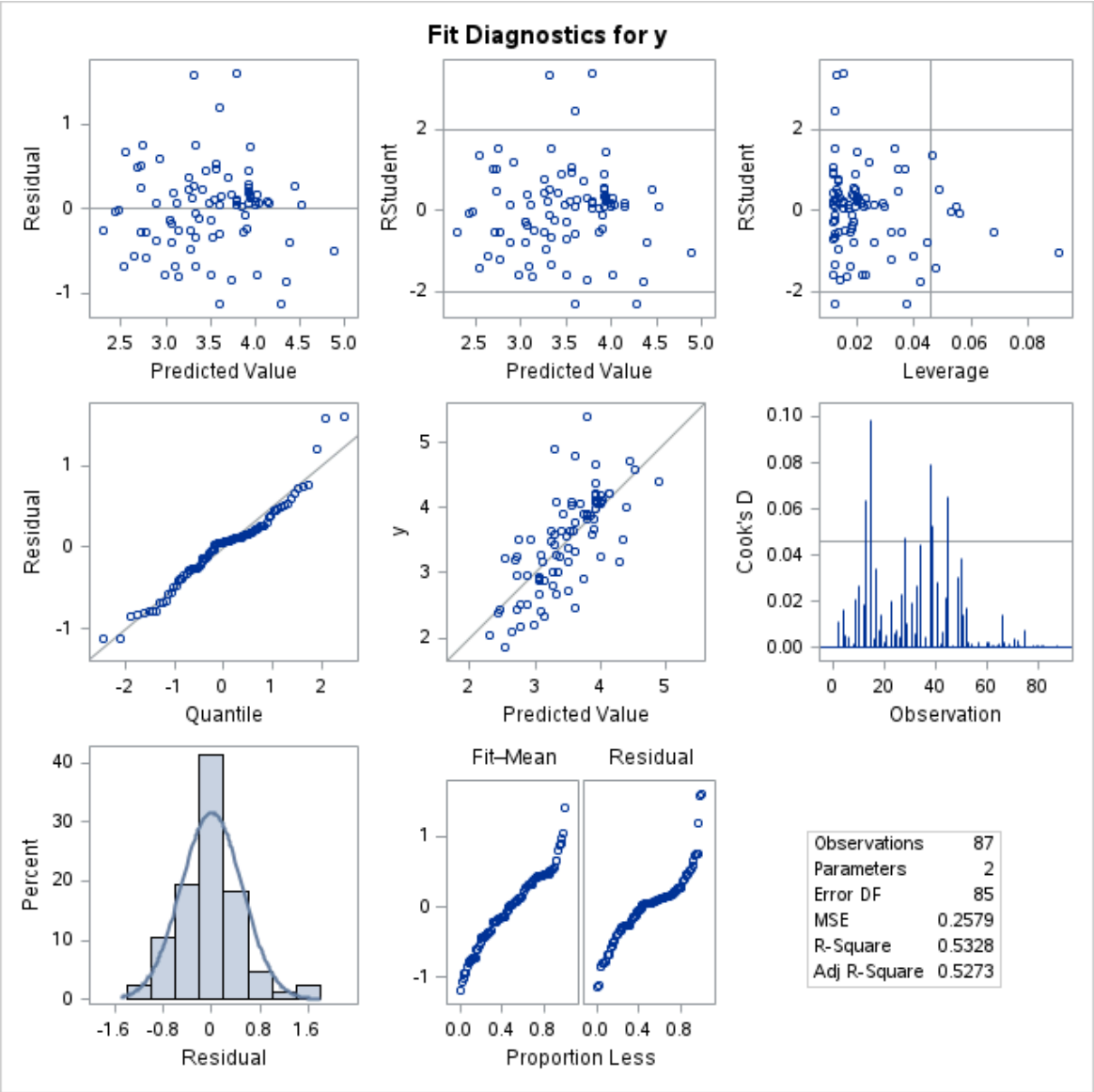
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	25.00179	25.00179	96.95	<.0001
Error	85	21.92034	0.25789		
Corrected Total	86	46.92213			

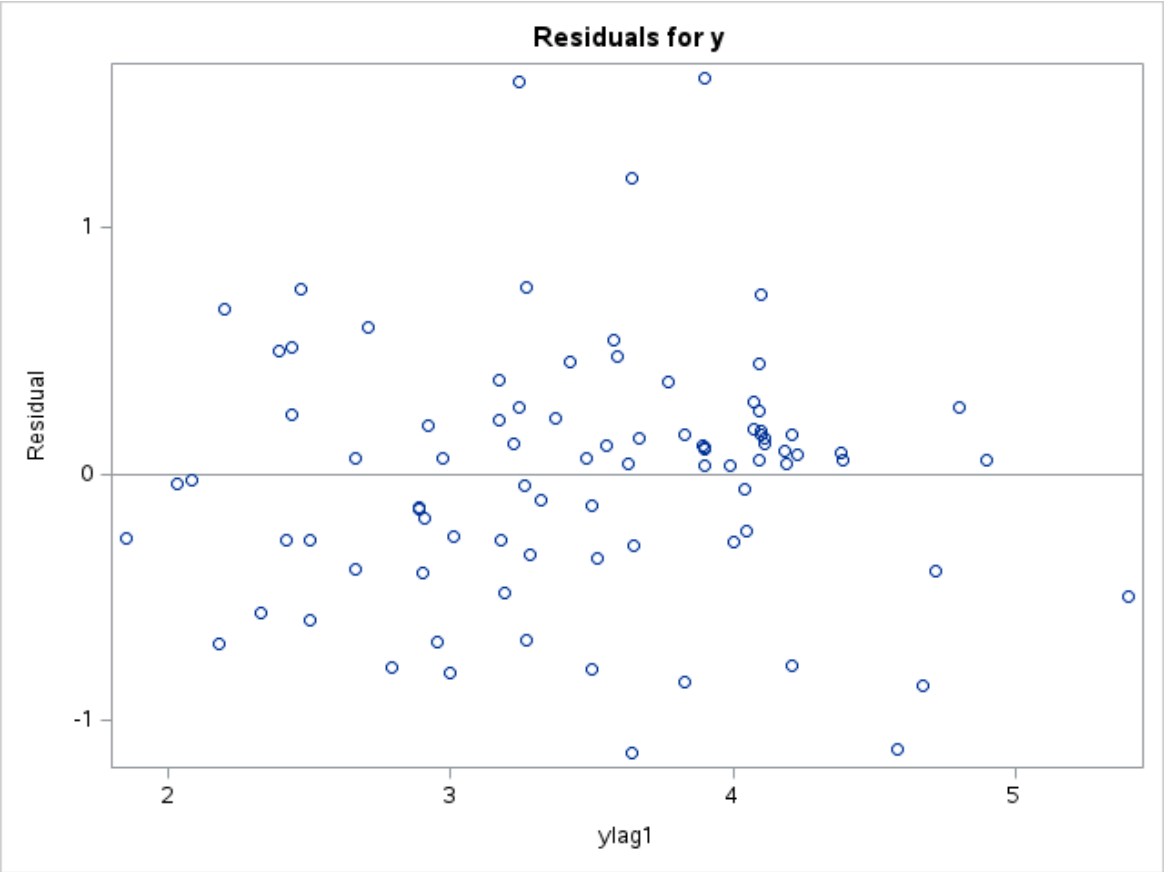
Root MSE	0.50783	R-Square	0.5328
Dependent Mean	3.48333	Adj R-Sq	0.5273
Coeff Var	14.57871		

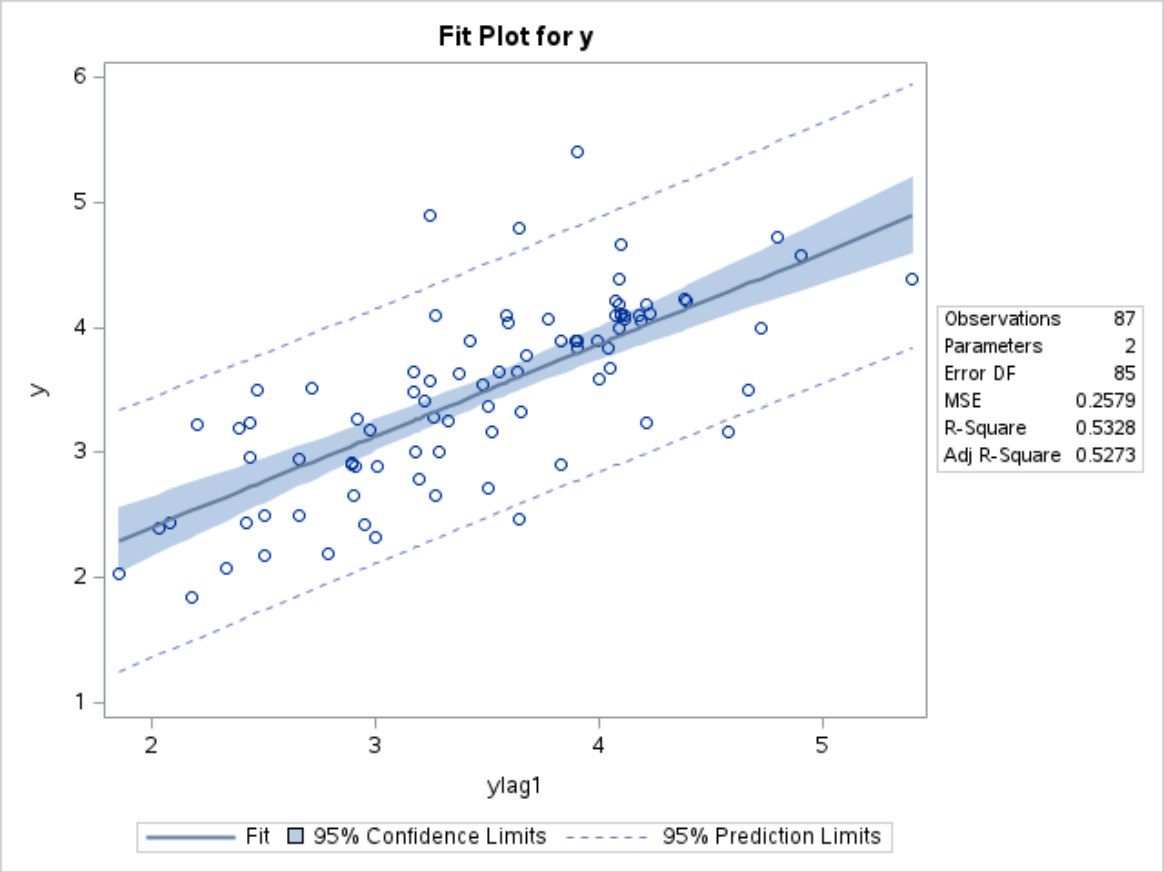
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	0.94249	0.26373	3.57	0.0006
ylag1	1	0.73092	0.07423	9.85	<.0001

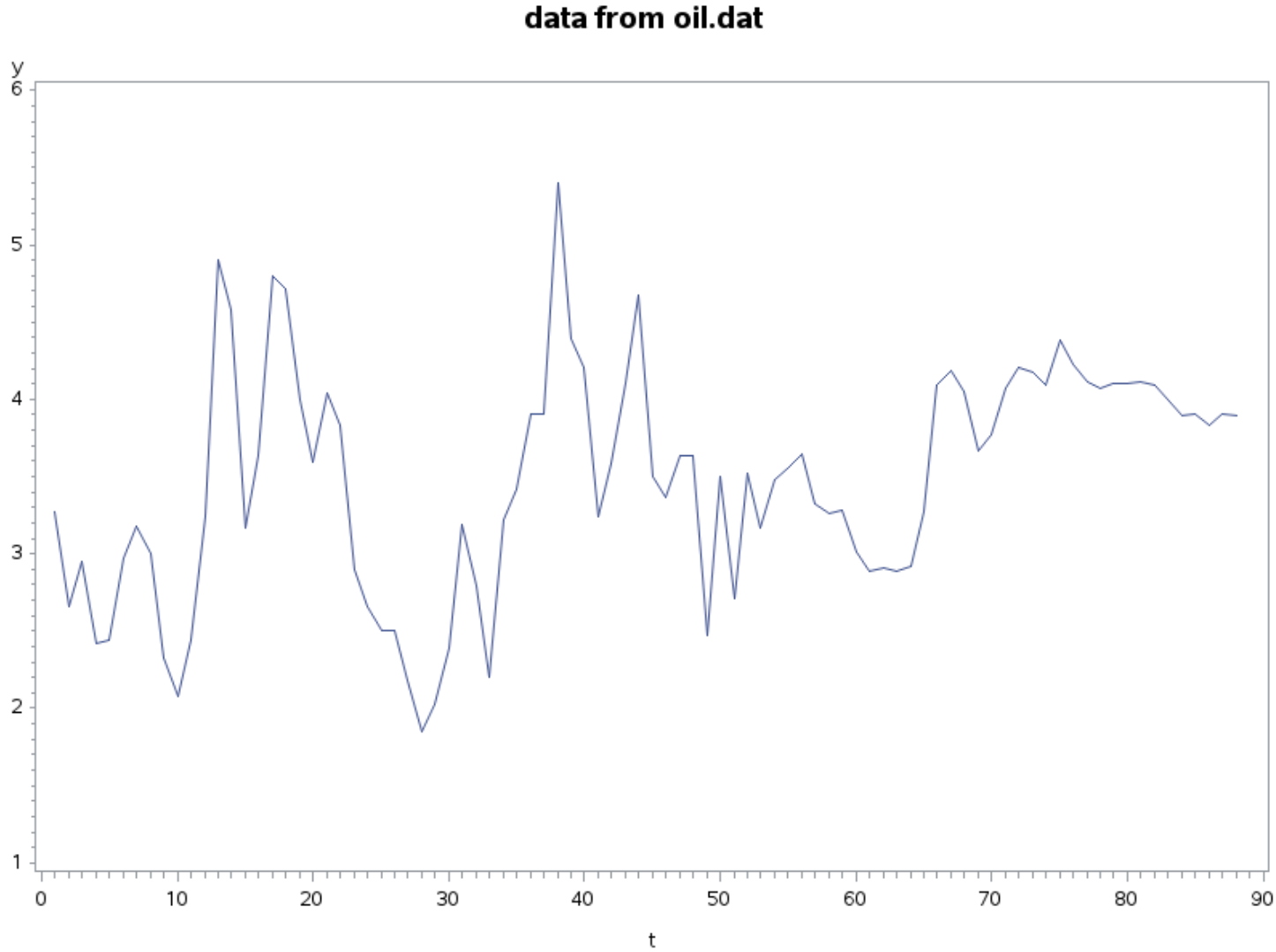
data from oil.dat

The REG Procedure
Model: MODEL1
Dependent Variable: y


Fit Diagnostics for y
Residual
Predicted Value
RStudent
Leverage
Quantile
y
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 87
Parameters 2
Error DF 85
MSE 0.2579
R-Square 0.5328
Adj R-Square 0.5273


Residuals for y
Residual
1
0
-1
2
3
4
5
ylag1


Fit Plot for y
y
6
5
4
3
2
1
2
3
4
5
ylag1
Observations 87
Parameters 2
Error DF 85
MSE 0.2579
R-Square 0.5328
Adj R-Square 0.5273
Fit
95% Confidence Limits
95% Prediction Limits



data from oil.dat

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	88
Number of Observations Used	86
Number of Observations with Missing Values	2

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	24.82804	12.41402	48.13	<.0001
Error	83	21.40833	0.25793		
Corrected Total	85	46.23637			

Root MSE	0.50787	R-Square	0.5370
Dependent Mean	3.49291	Adj R-Sq	0.5258
Coeff Var	14.54003		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1.00742	0.28315	3.56	0.0006

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
ylag1	1	0.76409	0.10850	7.04	<.0001
ylag2	1	-0.04968	0.10866	-0.46	0.6487

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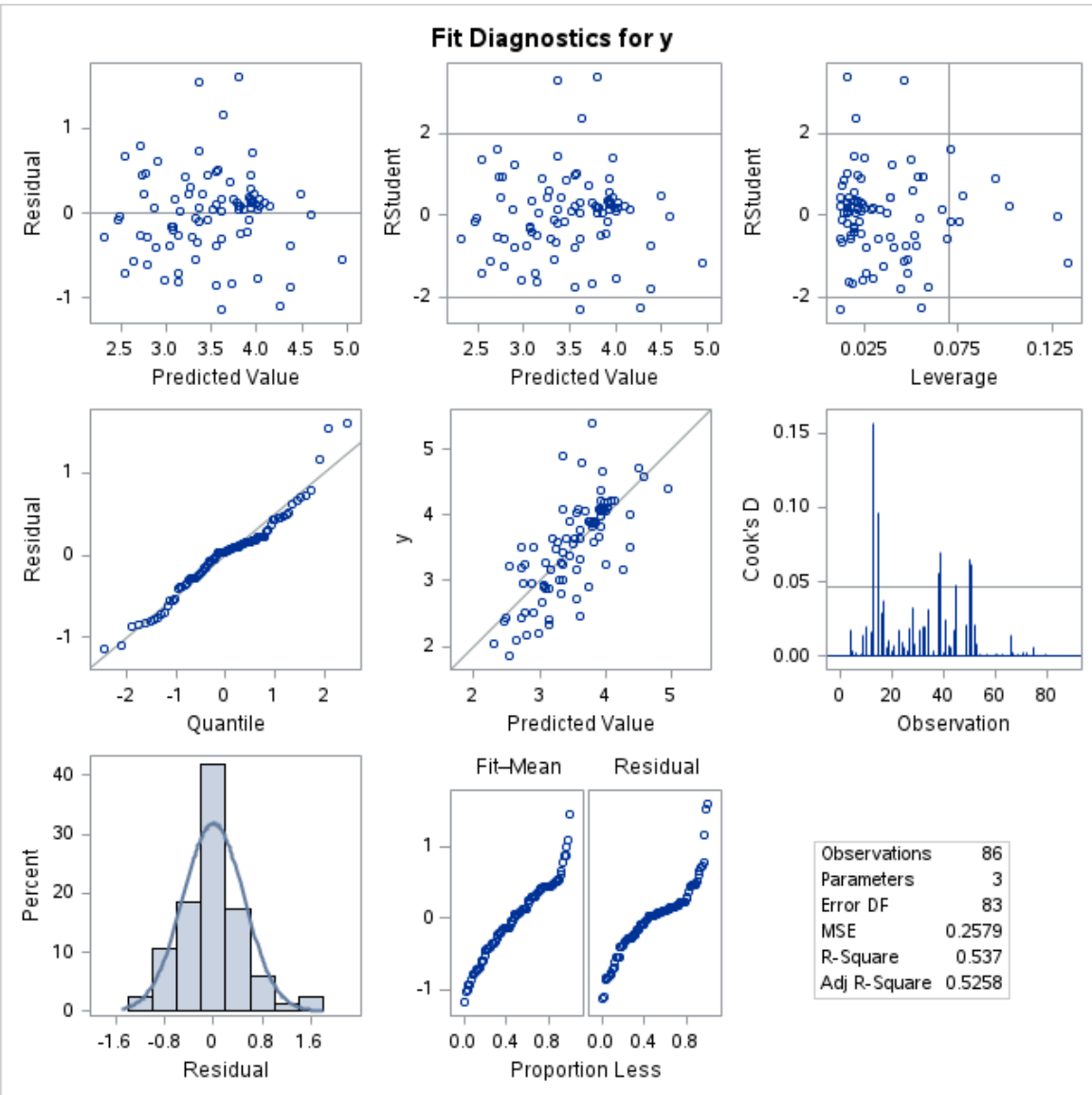
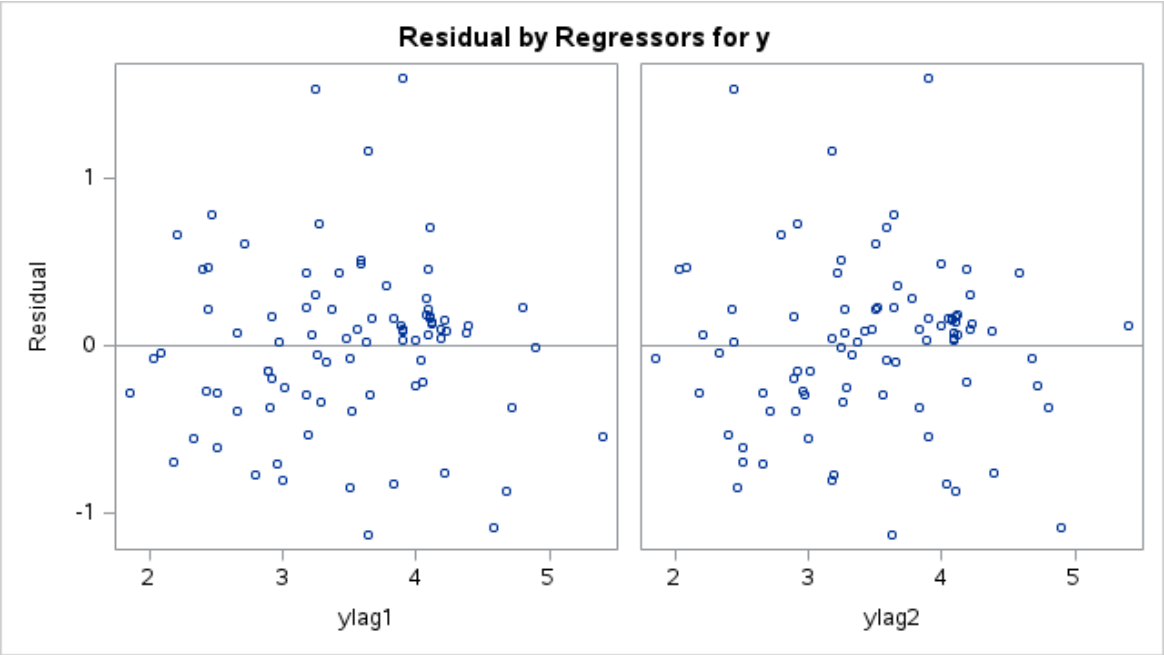
data from oil.dat

The REG Procedure
Model: MODEL1
Dependent Variable: y



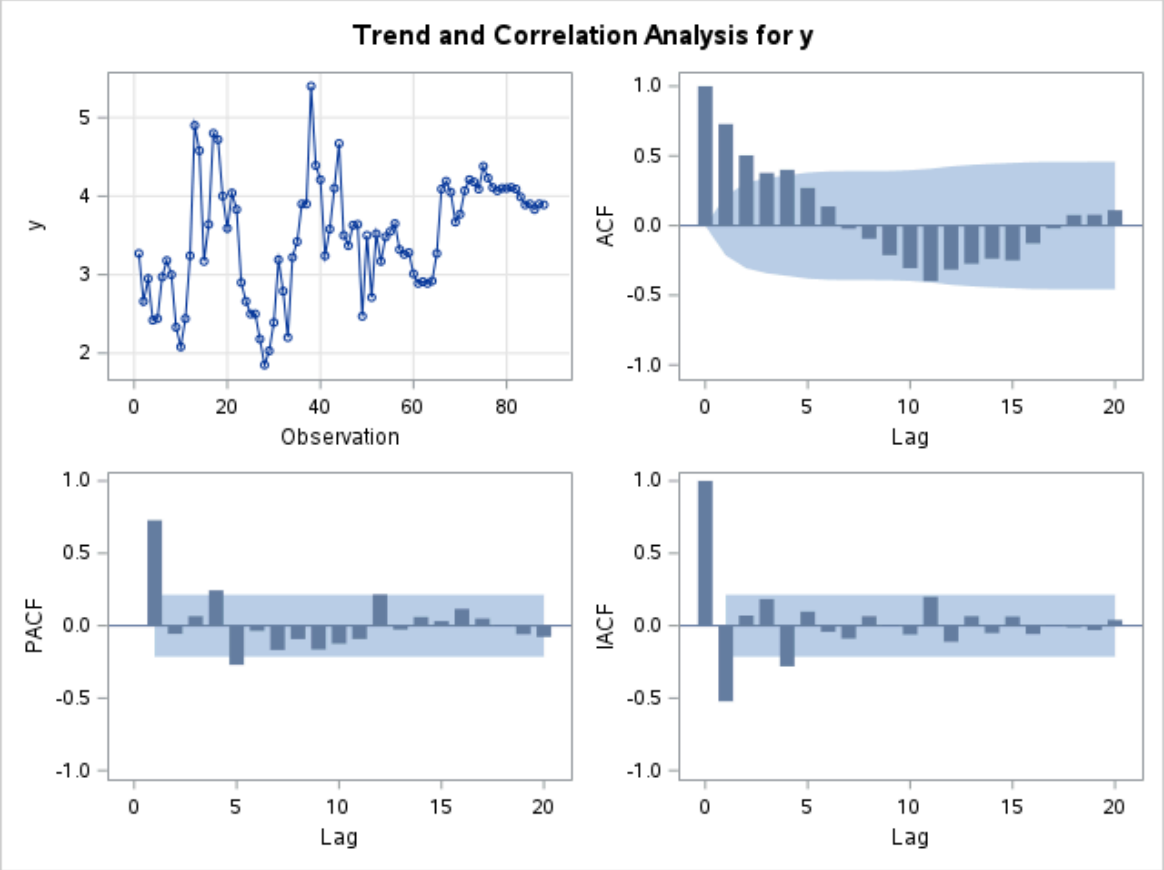


data from oil.dat

The ARIMA Procedure

Name of Variable = y	
Mean of Working Series	3.480909
Standard Deviation	0.73056
Number of Observations	88

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	108.85	6	<.0001	0.728	0.504	0.378	0.400	0.270	0.137
12	150.42	12	<.0001	-0.022	-0.094	-0.212	-0.305	-0.398	-0.315
18	173.59	18	<.0001	-0.274	-0.237	-0.249	-0.128	-0.020	0.076



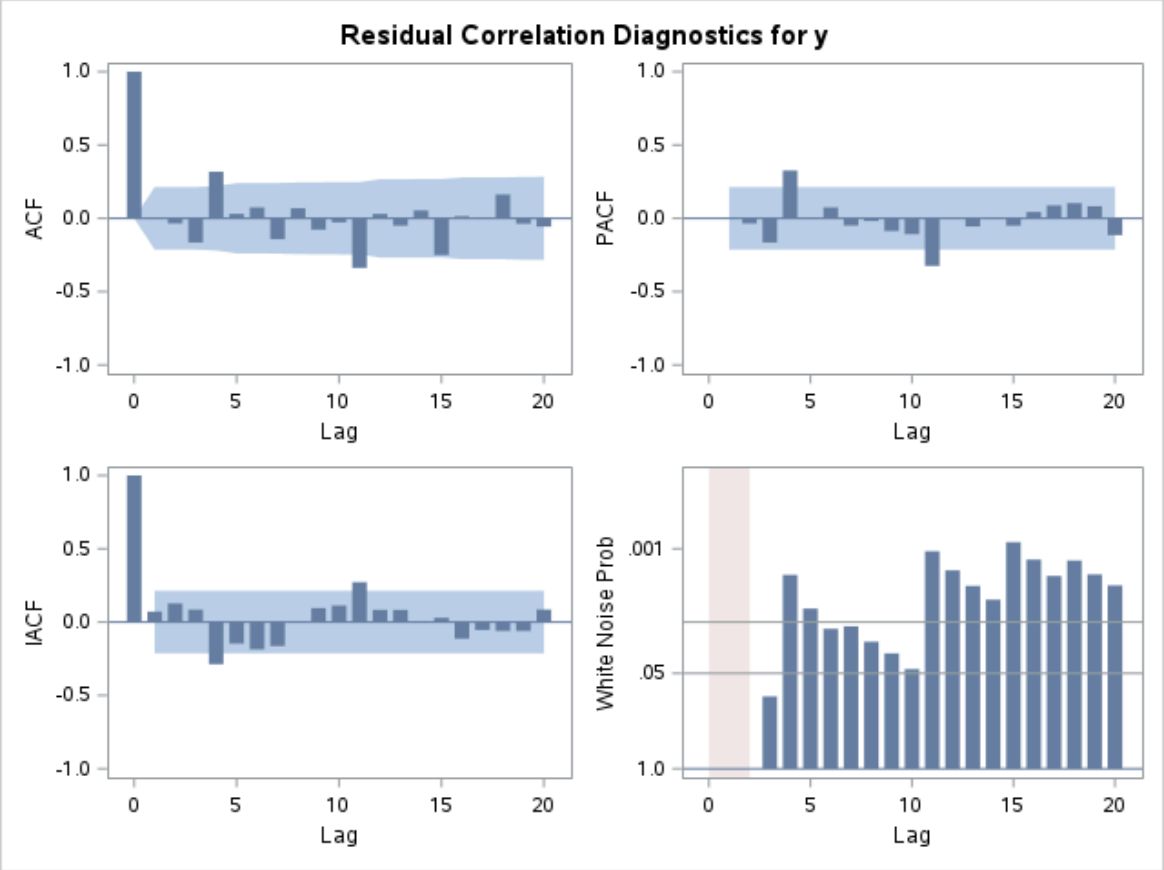
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	3.47610	0.17962	19.35	<.0001	0
AR1,1	0.77061	0.10833	7.11	<.0001	1
AR1,2	-0.05458	0.10854	-0.50	0.6164	2

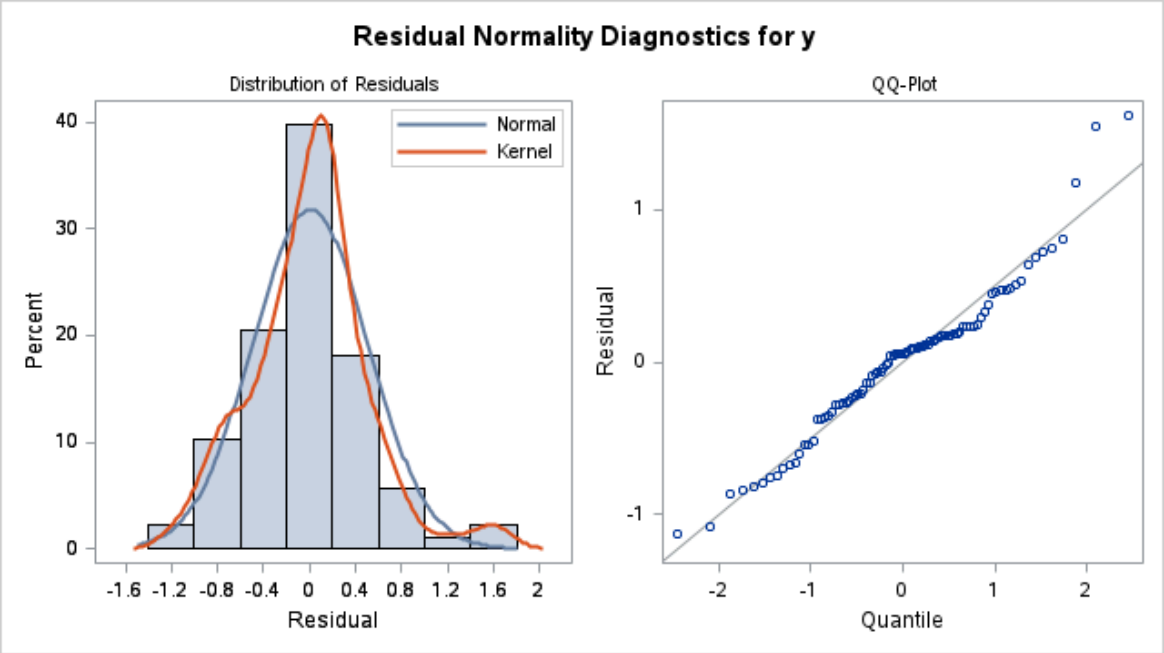
Constant Estimate	0.98709
Variance Estimate	0.257672
Std Error Estimate	0.507614
AIC	133.3469
SBC	140.7789
Number of Residuals	88

*** AIC and SBC do not include log determinant.**

Correlations of Parameter Estimates			
Parameter	MU	AR1,1	AR1,2
MU	1.000	-0.008	0.010
AR1,1	-0.008	1.000	-0.729
AR1,2	0.010	-0.729	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	12.81	4	0.0123	0.004	-0.035	-0.165	0.318	0.030	0.074
12	27.74	10	0.0020	-0.142	0.069	-0.077	-0.027	-0.338	0.030
18	38.08	16	0.0015	-0.050	0.053	-0.250	0.015	0.005	0.162
24	41.44	22	0.0073	-0.037	-0.057	0.132	0.071	0.030	0.020





Model for variable y	
Estimated Mean	3.476103

Autoregressive Factors	
Factor 1:	1 - 0.77061 B**(1) + 0.05458 B**(2)

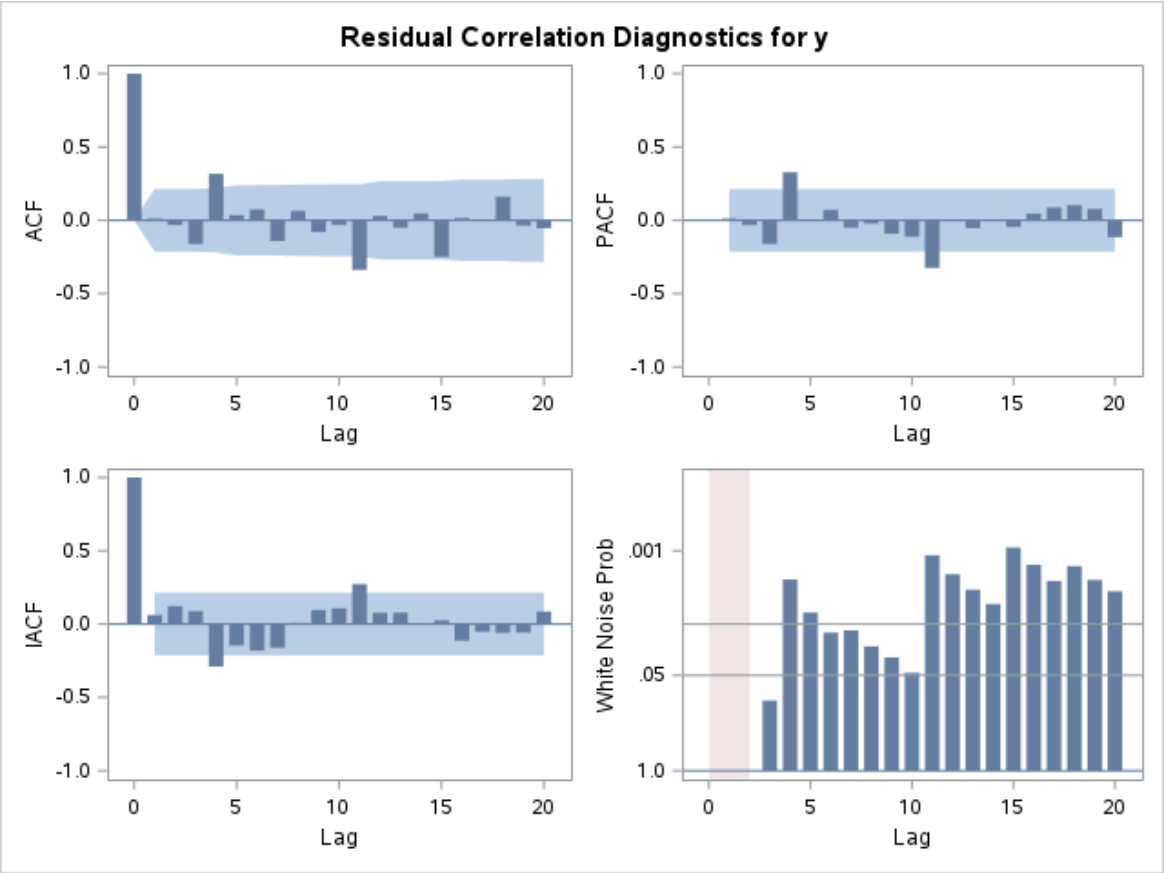
Maximum Likelihood Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	3.48690	0.18123	19.24	<.0001	0

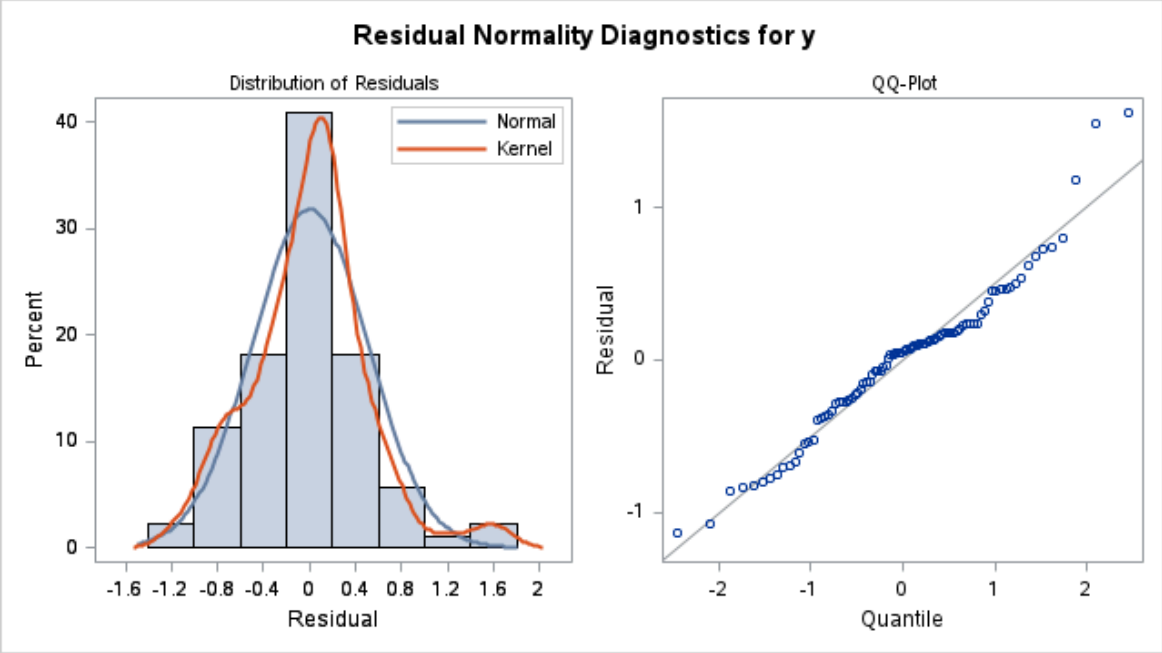
Maximum Likelihood Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
AR1,1	0.76008	0.10789	7.04	<.0001	1
AR1,2	-0.05002	0.10811	-0.46	0.6436	2

Constant Estimate	1.010983
Variance Estimate	0.257536
Std Error Estimate	0.50748
AIC	134.0479
SBC	141.4799
Number of Residuals	88

Correlations of Parameter Estimates			
Parameter	MU	AR1,1	AR1,2
MU	1.000	0.002	0.017
AR1,1	0.002	1.000	-0.729
AR1,2	0.017	-0.729	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	12.67	4	0.0130	0.012	-0.032	-0.162	0.317	0.035	0.074
12	27.59	10	0.0021	-0.141	0.064	-0.080	-0.032	-0.338	0.030
18	37.77	16	0.0016	-0.051	0.047	-0.249	0.016	0.007	0.161
24	41.12	22	0.0080	-0.037	-0.053	0.133	0.071	0.031	0.021

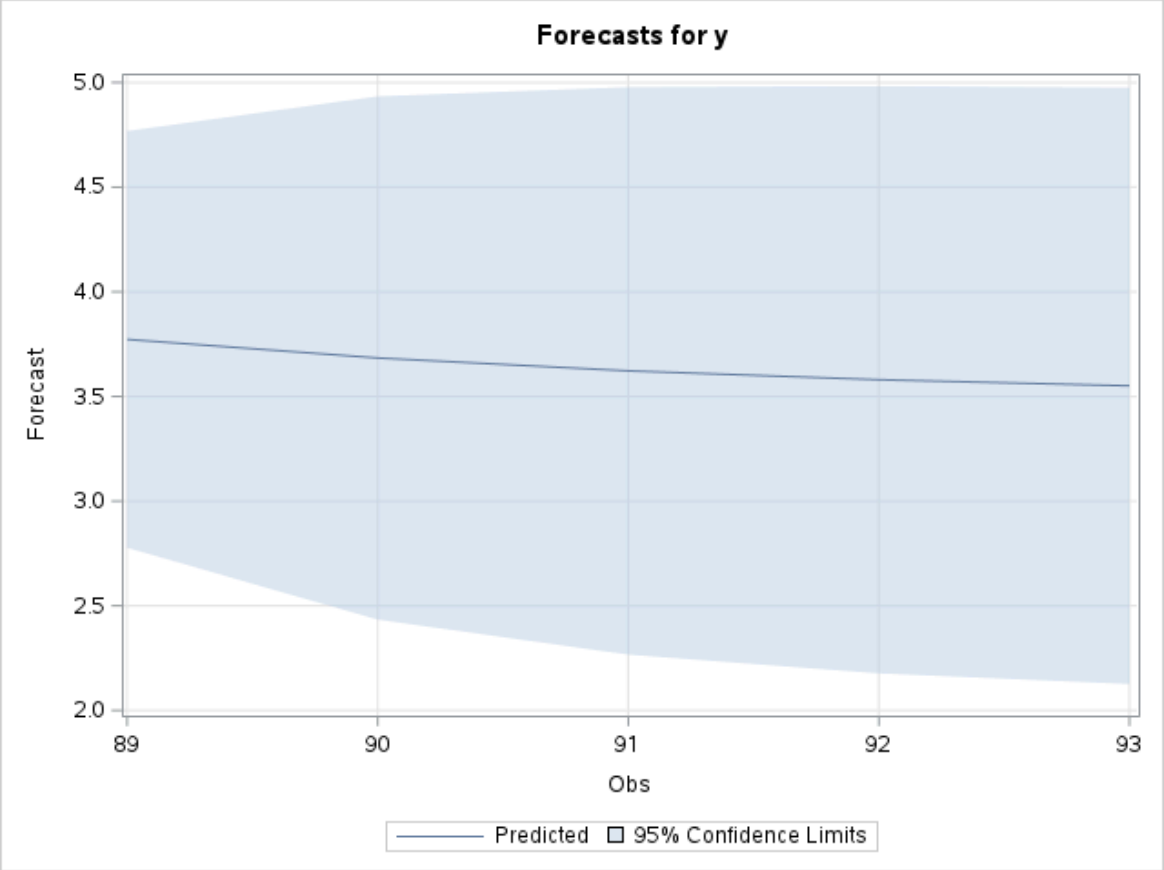




Model for variable y	
Estimated Mean	3.486903

Autoregressive Factors	
Factor 1:	1 - 0.76008 B**(1) + 0.05002 B**(2)

Forecasts for variable y				
Obs	**Forecast**	**Std Error**	**95% Confidence Limits**	
89	3.7726	0.5075	2.7780	4.7673
90	3.6839	0.6374	2.4346	4.9333
91	3.6224	0.6914	2.2672	4.9775
92	3.5800	0.7155	2.1776	4.9824
93	3.5509	0.7267	2.1267	4.9751

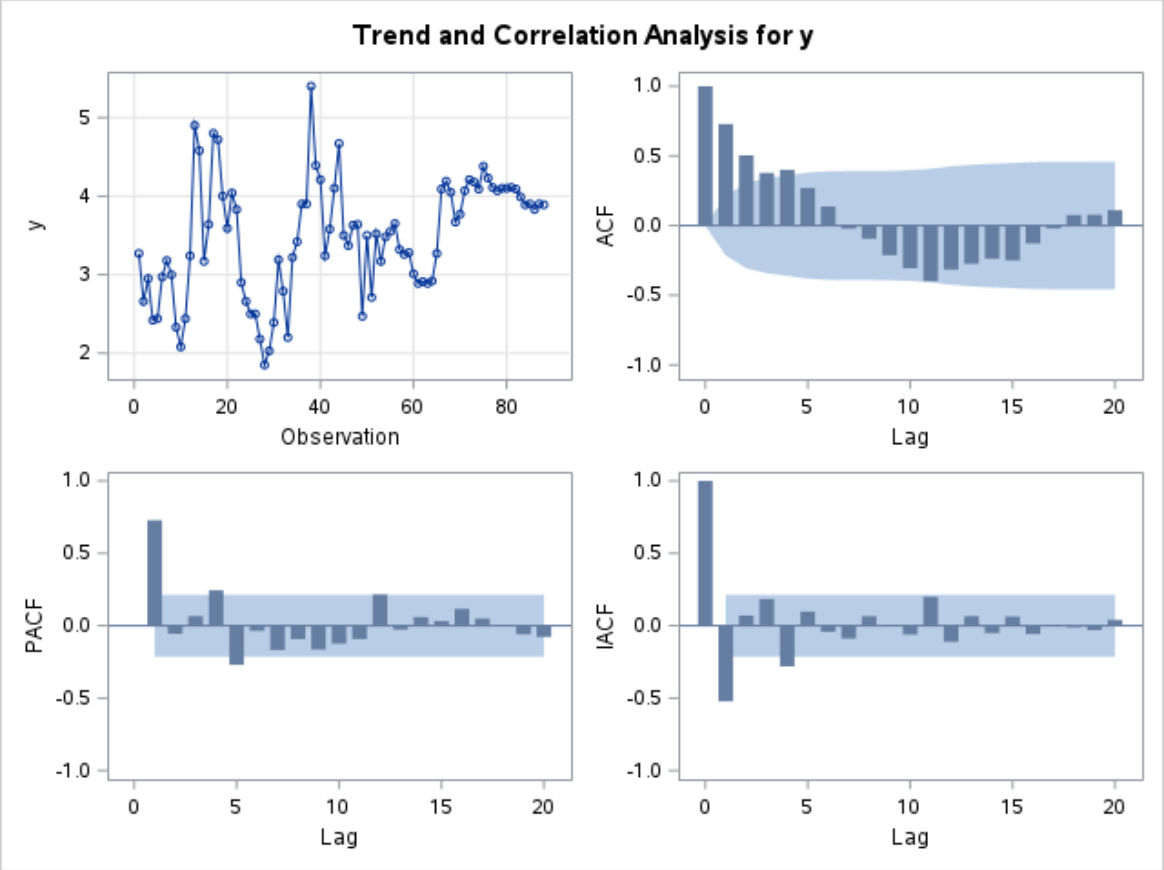


data from oil.dat

The ARIMA Procedure

Name of Variable = y	
Mean of Working Series	3.480909
Standard Deviation	0.73056
Number of Observations	88

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	108.85	6	<.0001	0.728	0.504	0.378	0.400	0.270	0.137
12	150.42	12	<.0001	-0.022	-0.094	-0.212	-0.305	-0.398	-0.315
18	173.59	18	<.0001	-0.274	-0.237	-0.249	-0.128	-0.020	0.076



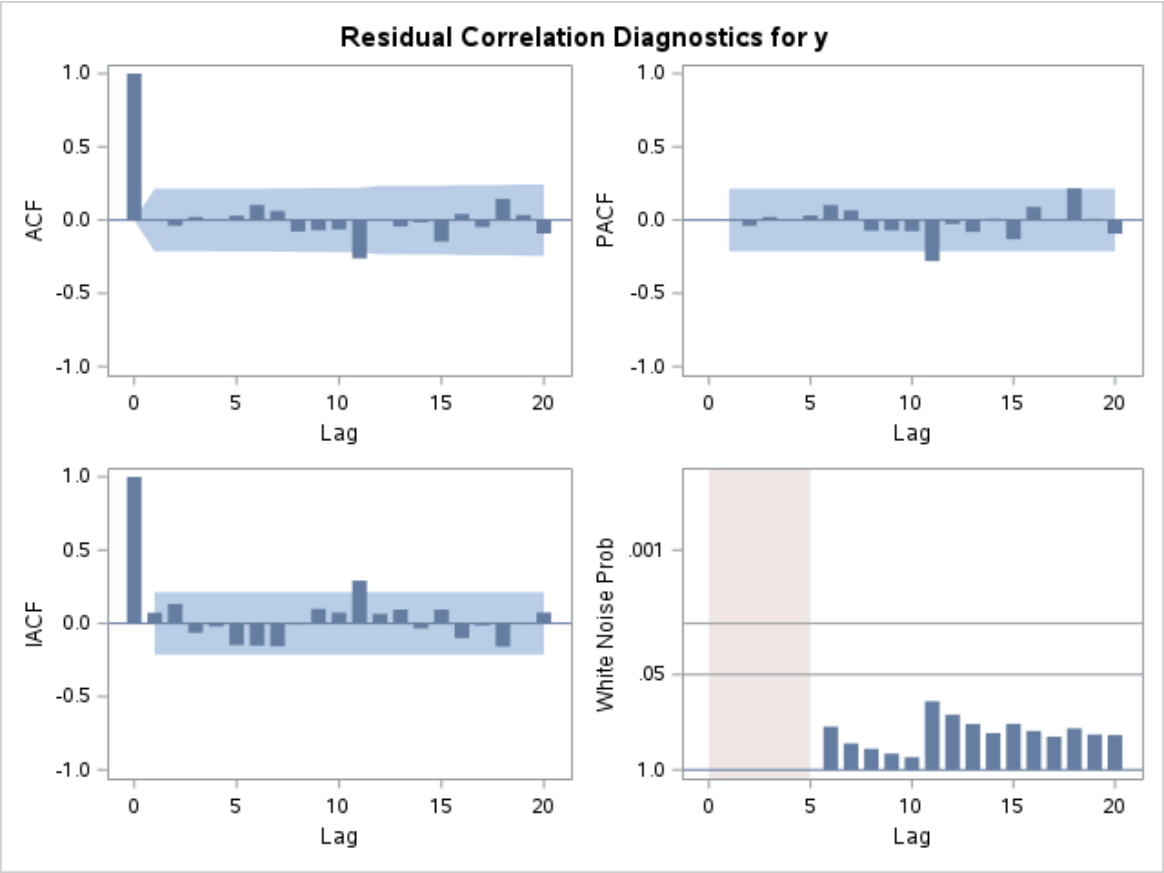
Maximum Likelihood Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	3.49005	0.19166	18.21	<.0001	0
AR1,1	0.81493	0.10482	7.77	<.0001	1
AR1,2	-0.10344	0.12711	-0.81	0.4158	2
AR1,3	-0.15600	0.12683	-1.23	0.2187	3
AR1,4	0.45207	0.12831	3.52	0.0004	4
AR1,5	-0.26537	0.10556	-2.51	0.0119	5

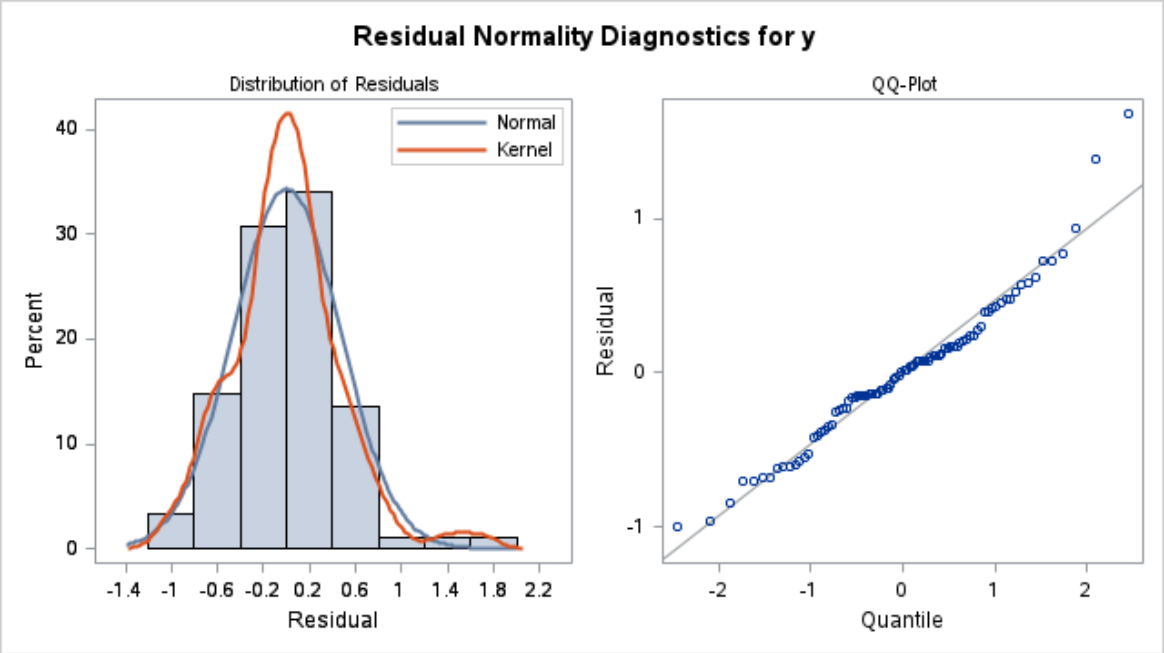
Constant Estimate	0.899745
Variance Estimate	0.229603
Std Error Estimate	0.479169
AIC	127.4237
SBC	142.2877
Number of Residuals	88

Correlations of Parameter Estimates						
Parameter	MU	AR1,1	AR1,2	AR1,3	AR1,4	AR1,5
MU	1.000	-0.007	0.011	-0.001	0.007	0.025
AR1,1	-0.007	1.000	-0.610	0.041	0.126	-0.263
AR1,2	0.011	-0.610	1.000	-0.501	-0.027	0.120
AR1,3	-0.001	0.041	-0.501	1.000	-0.509	0.055
AR1,4	0.007	0.126	-0.027	-0.509	1.000	-0.613
AR1,5	0.025	-0.263	0.120	0.055	-0.613	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	1.28	1	0.2570	0.002	-0.039	0.021	-0.001	0.030	0.103
12	10.21	7	0.1768	0.062	-0.079	-0.069	-0.064	-0.262	0.004

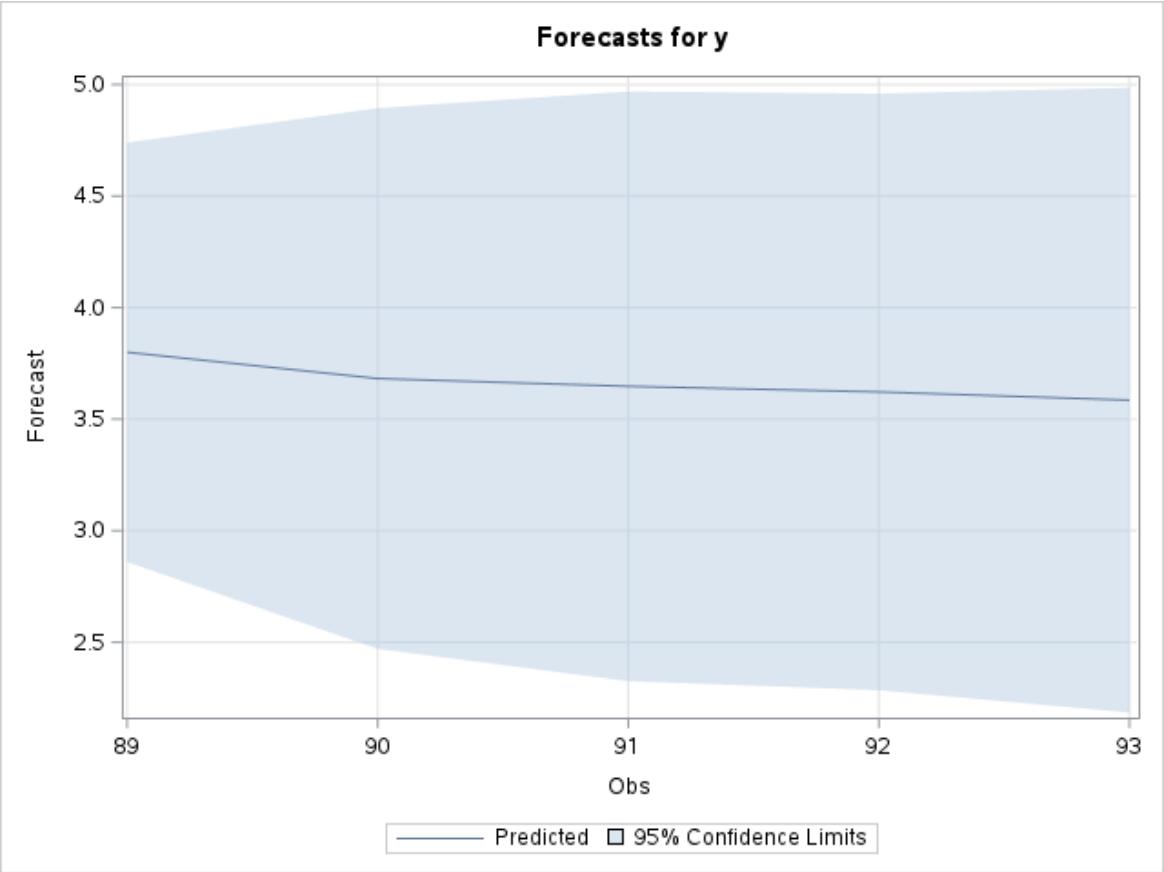
Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
18	15.59	13	0.2721	-0.044	-0.015	-0.147	0.041	-0.048	0.145
24	20.82	19	0.3470	0.035	-0.092	0.162	-0.002	0.055	0.072





Model for variable y	
Estimated Mean	3.490047

Autoregressive Factors

Autoregressive Factors	
Factor 1:	1 - 0.81493 B**(1) + 0.10344 B**(2) + 0.156 B**(3) - 0.45207 B**(4) + 0.26537 B**(5)

Forecasts for variable y				
Obs	**Forecast**	**Std Error**	**95% Confidence Limits**	
89	3.7997	0.4792	2.8606	4.7389
90	3.6820	0.6181	2.4705	4.8935
91	3.6472	0.6740	2.3262	4.9682
92	3.6219	0.6819	2.2854	4.9585
93	3.5852	0.7143	2.1852	4.9852